

PROFESSIONAL SUMMARY

- Data Engineer with 7+ years of experience designing and implementing scalable **ETL pipelines**, **data lakes**, and **cloud-native analytics solutions** across **AWS**, **Azure**, and **GCP**.
- Extensive hands-on expertise with **AWS (S3, Glue, Redshift, Lambda, Kinesis, MSK)** for building high-performance batch and streaming data workflows.
- Proficient in **Azure Data Factory (ADF)**, **Databricks (PySpark, Delta Lake, Unity Catalog)**, **Synapse Analytics**, and **ADLS**, supporting large-scale migrations and operational analytics.
- Skilled in **GCP (BigQuery, Dataflow, Dataproc, Pub/Sub)** for cost-effective real-time ingestion and analytics, and integrating **Apache Beam** for time-series transformations.
- Strong experience with **big data and orchestration tools** such as **Apache Spark, PySpark, Hadoop (HDFS, Hive), Kafka, Airflow, Matillion, Delta Lake, dbt**, and **Informatica** for processing complex, high-volume datasets.
- Delivered **AI/ML and GenAI solutions** using **Azure OpenAI, Hugging Face Transformers, LangChain, GPT-4 APIs, SageMaker, Spark ML, XGBoost, and MLflow**, supporting predictive analytics, anomaly detection, and intelligent automation.
- Experienced in developing and deploying infrastructure solutions using **Terraform**, with working knowledge of **CloudFormation** and **CDK**.
- Familiar with **network architecture principles**, including **VPNs, firewalls, routing, and security protocols** for cloud-hosted environments.
- Proficient in **SQL and NoSQL databases** including **Snowflake, Redshift, BigQuery, Synapse, PostgreSQL, DynamoDB, and Cassandra**, with strong query optimization and data modeling skills (Star/Snowflake Schema).
- Built **interactive dashboards** and advanced KPIs using **Power BI, Tableau, Looker, and QuickSight**, delivering actionable insights to business and executive teams.
- Experienced in **data governance and compliance** using **Azure Purview, IBM IGC, Ataccama DQ, Key Vault, Row-Level Security (RLS)**, and **Data Sensitivity Labels**, ensuring **GDPR, HIPAA, HITRUST, and SOC 2** compliance.
- Skilled in **CI/CD and infrastructure automation** with **Terraform, Jenkins, GitLab CI/CD, GitHub Actions, and Docker**, enabling seamless provisioning and deployment of pipelines and ML workflows.
- Adept at **observability and monitoring** using **DataDog, Grafana, Azure Monitor, and CloudWatch**, building proactive alerting, lineage tracking, and SLA enforcement for mission-critical workloads.
- Collaborates closely with cross-functional teams, including **AI/ML, DevOps, Bioinformatics**, and Business stakeholders, to deliver enterprise-scale, multi-cloud-ready analytics and AI-driven solutions.

EDUCATION

- Masters in Information Science - Trine University, MI.

CERTIFICATIONS

- **Microsoft Certified:** Azure Data Fundamentals
- **AWS Certified:** DevOps Engineer
- **Google Cloud Certified:** Cloud Engineer

TECHNICAL SKILLS

- **Cloud Platforms & Services:** AWS (S3, Glue, Redshift, Lambda, Kinesis, MSK, EMR), Azure (Data Factory, Databricks, Synapse, ADLS, Purview, Event Hubs, Event Grid, Cosmos DB, API Management), GCP (BigQuery, Dataflow, Dataproc, Pub/Sub), Snowflake, Palantir Foundry
- **Big Data & ETL Processing:** Spark, PySpark, Scala, Matillion, Hadoop (HDFS, Hive), Kafka, Airflow, Delta Lake, dbt, Apache Beam, Informatica, AWS Step Functions
- **Databases & Data Warehousing:** Snowflake, Redshift, BigQuery, Synapse, PostgreSQL, MongoDB, Couchbase, DynamoDB, Cassandra
- **Streaming & Real-Time Analytics:** Kinesis, Spark Streaming, Azure Stream Analytics, Pub/Sub, Flink, Apache Pulsar, Materialize
- **DataOps & Infrastructure:** Terraform, Jenkins, Docker, Kubernetes, GitHub CI/CD, GitHub Actions
- **BI, Visualization & Monitoring:** Power BI, Tableau, Looker, QuickSight, DataDog, Grafana
- **Programming & Data Modeling:** Python, SQL, PySpark, Scala, Java, SQL Mesh, GeoPandas, Shapely
- **Networking & Security:** VPNs, TCP/IP, IAM, RBAC, Azure Key Vault, Firewalls
- **LLMs & GenAI:** Azure OpenAI, OpenAI API, Hugging Face Transformers, LangChain, Prompt Engineering, BERT, GPT-2, GPT-3.5/4
- **ML & Advanced Analytics:** Azure ML, SageMaker, Spark MLlib, XGBoost, Prophet, scikit-learn
- **Data Governance & Quality:** Azure Purview, Ataccama DQ, IBM IGC, Great Expectations, Data Sensitivity Labels, Row-Level Security (RLS)

PROFESSIONAL EXPERIENCE

American Airlines, Dallas, TX | Aug 2024 – Present

Azure Data Engineer

Responsibilities:

- Designed and deployed enterprise-grade, end-to-end ETL pipelines using **Azure Data Factory (ADF)**, **Databricks (PySpark, Delta Lake, Unity Catalog)**, **Snowflake**, and **Synapse Analytics**, integrating **structured, semi-structured, streaming, and API-based**

datasets (Kafka, Event Hubs, REST APIs) to support real-time and batch processing across flight operations, customer, and finance domains.

- Implemented a medallion architecture (**Bronze, Silver, Gold layers**) within **Databricks** and **Delta Lake**, transforming raw streaming and transactional datasets into curated and aggregated layers optimized for predictive analytics, **ML** model consumption, and operational reporting.
- Developed and optimized complex **T-SQL** and **Snowflake SQL** procedures, window functions, materialized views, and dynamic pivot queries, improving BI dashboard performance for mission-critical reports used by revenue management and operations teams.
- Integrated **Generative AI (GenAI)** and **LLM pipelines** using **Azure OpenAI**, **LangChain**, **Hugging Face Transformers**, and **GPT-4 APIs** for automated anomaly detection, natural language summarization of operational metrics, and intelligent data classification.
- Built and trained **machine learning models** using **Azure ML**, **Spark MLlib**, and **scikit-learn** for flight delay prediction, revenue forecasting, and customer behavior analysis, leveraging **Databricks MLflow** for experiment tracking and automated deployment pipelines.
- Migrated multiple legacy **SSIS** and on-prem **SQL Server ETL** processes into cloud-native solutions using **ADF**, **Databricks**, and **Snowflake**.
- Designed and optimized **data models, schema, and indexing strategies** for **MongoDB** and **AWS DynamoDB**, ensuring low-latency queries and scalability for real-time applications.
- Implemented **sharding, replication, and partitioning strategies** to support distributed data workloads and high availability.
- Collaborated with DevOps teams to automate **backup, recovery, and HA/DR strategies** using **Terraform** and **Ansible**, enhancing fault tolerance.
- Integrated **Kafka streaming** with **NoSQL** databases for real-time analytics and event-driven data processing.
- Automated data quality, validation, and observability using **dbt**, **Great Expectations**, **Azure Monitor**, and **DataDog**, establishing alerting, lineage, and SLA enforcement across mission-critical pipelines.
- Created self-service **Snowflake data marts** for Finance, Operations, and Marketing, implementing role-based access, time-travel, and data sharing features, enabling faster insights for analysts and data scientists.
- Developed interactive **Power BI** dashboards with **DAX** measures and **DirectQuery** connections to **Snowflake/Synapse**, delivering real-time KPI monitoring and predictive trend analysis for senior executives.
- Applied data security, governance, and compliance frameworks using **Azure Purview**, **Key Vault**, **Private Endpoints**, **Row-Level Security (RLS)**, and **Dynamic Data Masking**, ensuring GDPR, CCPA, and HIPAA compliance for sensitive operational datasets.
- Designed CI/CD workflows using **Terraform** and **GitHub Actions** for provisioning infrastructure on **multi-cloud platforms**, applying DevOps best practices and cost-optimization strategies.
- Built fully automated **CI/CD workflows** using **Jenkins**, **GitHub Actions**, **Terraform**, and **Docker** for provisioning **ADF pipelines**, **Databricks clusters**, and **Snowflake resources**.
- Collaborated with **DevOps**, **AI/ML teams**, and business stakeholders to design scalable, multi-cloud-ready architectures, supporting streaming analytics, **ML pipelines**, and **GenAI-driven data solutions** for enterprise decision-making.

Mylon Laboratories, India | Aug 2020 – Jul 2023

Azure Data Engineer

Responsibilities:

- Designed and implemented high-throughput ETL and ELT pipelines using **Matillion**, **Apache NiFi**, and **Azure Data Factory (ADF)** to integrate **EHR datasets (HL7/FHIR)**, claims feeds, and external provider networks into a **HIPAA-compliant Azure Data Lakehouse** built on **ADLS Gen2** and **Snowflake**.
- Architected real-time streaming pipelines using **Azure Event Hubs**, **Event Grid**, and **Stream Analytics**, enabling ingestion of telehealth, **IoT device**, and **pharmacy feeds** into **Databricks Delta tables** for downstream care analytics.
- Leveraged **Azure Dataverse** and **Cosmos DB** for handling **semi-structured, JSON, and document-based medical records**, enabling care teams to query unstructured data at scale with minimal latency.
- Developed multi-layered **star and snowflake schemas** optimized for **population health analytics and predictive modeling**, enabling actuarial teams to run risk scoring and utilization forecasting with high performance.
- Automated data validation, PHI masking, and quality scoring using **Azure Logic Apps**, **Power Automate**, **dbt**, and **Great Expectations**, providing **end-to-end lineage and SLA reporting** for compliance audits.
- Built **ML** and statistical pipelines for fraud detection, readmission risk, and chronic disease progression modeling using **Azure ML**, **Spark MLlib**, **XGBoost**, and **Prophet**, integrating **MLflow** for tracking, deployment, and drift detection.
- Migrated critical datasets from **traditional RDBMS (SQL Server/Oracle)** to **Cassandra** and **Azure Cosmos DB**, improving horizontal scalability and reducing query response times by 40%.
- Integrated **IAM**, **private endpoints**, **VNet/Subnet configurations**, and encryption standards to enforce enterprise security and compliance.
- Tuned **NoSQL performance** via query optimization, data partitioning, and caching strategies using **Redis**.
- Developed automation scripts in **Python** and **Shell** for database health monitoring, capacity planning, and alerting mechanisms.
- Deployed **FHIR-compliant APIs** via **Azure API Management** to enable secure, scalable consumption of curated clinical datasets by provider and research teams.
- Created interactive **Power BI dashboards** with Direct Lake Mode and composite models to deliver claims cost insights, readmission KPIs, and predictive alerts directly to care managers and executives.
- Configured advanced governance and security using **Azure Purview**, **Key Vault**, **Private Endpoints**, **Data Sensitivity Labels**, and **Role-Based Access Control (RBAC)** to ensure compliance with **HIPAA**, **HITRUST**, and **SOC 2** standards.
- Migrated legacy **Informatica** and **Hadoop-based ETL pipelines** into cloud-native **Databricks** and **Snowflake workflows**, improving processing throughput and reducing operational overhead.
- Implemented **GitLab CI/CD** and **Terraform automation** to provision **ADF pipelines**, **Databricks clusters**, **Snowflake warehouses**, and **Event Hub** namespaces, streamlining deployments across environments.

- Collaborated with clinical data scientists, care management teams, and actuarial groups to deliver predictive analytics solutions that improved care gap closure rates and reduced claims leakage.

Rwartz Group, India | Sep 2018 – Jul 2020

AWS Data Engineer

Responsibilities:

- Designed and deployed enterprise-grade ETL pipelines using **Python, Scala, AWS Glue (Crawlers, Catalog, Jobs)**, and **Spark** to process large-scale genomic, clinical trial, and EHR datasets, integrating multi-terabyte feeds from lab systems, clinical partners, and public genomic repositories into **AWS S3 data lakes** for downstream analytics.
- Orchestrated streaming and batch ingestion using **Kafka Connect** and **AWS MSK**, enabling ingestion of **HL7/FHIR** messages, real-time trial results, and sensor data into **S3, Redshift Spectrum**, and **Snowflake**, ensuring low-latency data delivery for research teams.
- Built **PySpark** ETL frameworks in **AWS Glue** with dynamic **partitioning, bucketing, and windowed transformations**, applying custom serialization and predicate pushdown optimizations to cut processing times across clinical data pipelines.
- Developed **PostgreSQL, Redshift, and Snowflake** schemas with **materialized views, distribution keys, and clustering strategies** for efficient querying of multi-billion-row trial datasets, enabling faster exploratory research and regulatory reporting.
- Integrated **Informatica Cloud** for ingestion and transformation of **structured and semi-structured EHR (HL7, CCD, FHIR) data**, implementing schema drift detection, automated validation, and PHI masking, ensuring full **HIPAA and 21 CFR Part 11 compliance**.
- Implemented ontology-driven semantic models in **Palantir Foundry**, mapping clinical, genomic, and trial data to standardized ontologies, while leveraging lineage tracking and metadata versioning to streamline audits and regulatory submissions.
- Automated multi-stage workflows using **AWS Step Functions** to orchestrate **Glue ETL jobs, Redshift/Snowflake loads**, and Foundry data transformations, achieving resilient, event-driven orchestration across multi-environment pipelines.
- Created **PySpark**-based log and anomaly detection pipelines for schema validation, PHI exposure monitoring, and trial data consistency checks, integrating alerts via **AWS CloudWatch, SNS**, and **Slack** for rapid incident response.
- Integrated **IBM IGC (Information Governance Catalog)** with **Ataccama DQ** to enforce data quality rules, metadata lineage tracking, and PHI classification, improving regulatory transparency and reducing downstream errors.
- Developed fully automated **CI/CD pipelines** with **GitLab, Terraform, and Docker**, provisioning AWS Glue jobs, Redshift clusters, Palantir Foundry transformations, and IAM roles across multiple environments, reducing deployment cycles.
- Partnered with bioinformatics, data science, and compliance teams to deliver curated, governed data products that accelerated drug discovery timelines and improved clinical trial insights through advanced analytics and machine learning-ready datasets.

Siemens, India | Jun 2017 – Aug 2018

Data Engineer

Responsibilities:

- Designed and maintained **batch and streaming data pipelines** using **Databricks (PySpark, Delta Lake)**, **Hadoop (HDFS, Hive, Spark Streaming)**, and **Kafka** to process industrial IoT sensor telemetry, equipment logs, and ERP system data, enabling predictive maintenance and anomaly detection.
- Migrated legacy **Hadoop (HDFS/YARN)** clusters to **GCP Dataproc**, integrating with **Pub/Sub** for **streaming ingestion** and **BigQuery** for scalable analytics, achieving **25% cost savings** on infrastructure.
- Enhanced **Delta Lake** performance by enabling ACID transactions, schema evolution, and time-travel queries in **Databricks**, boosting **real-time data reliability** for operational dashboards.
- Developed **Airflow DAGs** to orchestrate **Spark/Hadoop ETL pipelines**, implementing **checkpointing, retry mechanisms**, and **Slack alerts** for **SLA adherence** and proactive error recovery.
- Created partitioned and clustered **BigQuery tables** with **materialized views** and **federated queries**, improving cost efficiency and query speeds for plant operations analytics by 30%.
- Incorporated **geospatial analytics** using **Python (GeoPandas, Shapely)** for equipment hotspot detection, enabling proactive interventions and reducing equipment downtime.
- Integrated **Grafana** and **DataDog** for **cluster health monitoring, ETL job metrics, and custom SLA alerts**, ensuring early detection of failures and resource optimization.
- Built streaming analytics dashboards by combining **Databricks Delta Live Tables** with **BigQuery BI Engine**, enabling near real-time tracking of **factory KPIs** for operations leadership.
- Leveraged **Apache Beam** within **Dataproc** to process **complex IoT time-series transformations**, improving downstream model readiness for predictive analytics teams.
- Implemented **CI/CD automation** with **Jenkins** and **Terraform** for **Dataproc clusters, Airflow DAGs, and Databricks jobs**, accelerating environment deployments by 40%.